```
Mozambique, step 2 (2026-11-02)

 chain     seed iterations.saved acceptance.rate final.log.likelihood best.log.likelihood errored.proposals
     1 20263016             5000           0.277             -39190.4            -39186.1                 0
     2 20263017             5000           0.277             -39151.3            -39145.0                 0
     3 20263018             5000           0.277             -39206.1            -39201.7                 0
     4 20263019             5000           0.275             -39223.1            -39210.6                 0
                                              file
 mcmc_mozambique_step2_chain1_2026-10-31.Rdata
 mcmc_mozambique_step2_chain2_2026-10-31.Rdata
 mcmc_mozambique_step2_chain3_2026-10-31.Rdata
 mcmc_mozambique_step2_chain4_2026-11-01.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):
     age.30.to.39.transmission.multiplier.1 over.80.mortality.intercept.multiplier.male
                                      2.658                                      2.024
    age.25.to.49.hiv.mortality.multiplier.1             hiv.specific.mortality.rates.0
                                      1.996                                      1.912
    age.15.to.24.hiv.mortality.multiplier.0     age.20.to.29.transmission.multiplier.0
                                      1.853                                      1.736
     age.0.to.14.hiv.mortality.multiplier.1             hiv.specific.mortality.rates.1
                                      1.735                                      1.719
           male.hiv.mortality.multiplier.0     age.0.to.14.hiv.mortality.multiplier.2
                                      1.713                                      1.664
     age.40.to.49.transmission.multiplier.1               age.20.to.24.base.aging.rate
                                      1.635                                      1.596
    age.25.to.49.hiv.mortality.multiplier.0     age.20.to.29.transmission.multiplier.2
                                      1.572                                      1.546
     age.20.to.29.transmission.multiplier.1     age.20.to.29.transmission.multiplier.3
                                      1.532                                      1.521
     age.30.to.39.transmission.multiplier.2     age.15.to.19.transmission.multiplier.1
                                      1.464                                      1.459
     age.15.to.19.transmission.multiplier.2                                 trate.0
                                      1.437                                      1.432
    age.15.to.24.hiv.mortality.multiplier.2     age.5.to.14.hiv.mortality.multiplier.0
                                      1.427                                      1.413
     age.30.to.39.transmission.multiplier.3                                 trate.2
                                      1.368                                      1.353
  age.50.and.over.transmission.multiplier.0
                                      1.349

Merged simset:
   location simulations total.weight                                          file
 Mozambique        1000         1000 simset_mozambique_step2_chains1234_2026-11-01.Rdata

Fit check (merged simset vs every calibration target, 2000-2025): 57 of 63 targets flagged (worst 30 shown; all in <location>_fit_check.csv)
     data.type         age    sex years.with.data pct.years.in.sim.95 median.sim.vs.data last.year last.sim.vs.data
 hiv.mortality       25-49 female              26                   0              x0.23      2025            x0.06
 hiv.mortality       15-24   male              26                   0              x3.39      2025            x1.56
 hiv.mortality 50 and over   male              26                   0              x3.05      2025            x2.20
 hiv.mortality       25-49   both              26                   0              x0.36      2025            x0.16
    engagement        0-14   both              11                   0          -40.2 pts      2025        -15.8 pts
 hiv.mortality 50 and over   both              26                   8              x2.05      2025            x1.25
 hiv.mortality       10-19   male              26                   0              x1.97      2025            x1.33
 hiv.mortality       25-49   male              26                   4              x0.51      2025            x0.29
 hiv.mortality       15-24   both              26                  38              x1.80      2025            x0.95
```

Mozambique step 2: likelihood

value

0
−10000
−20000
−30000
−40000

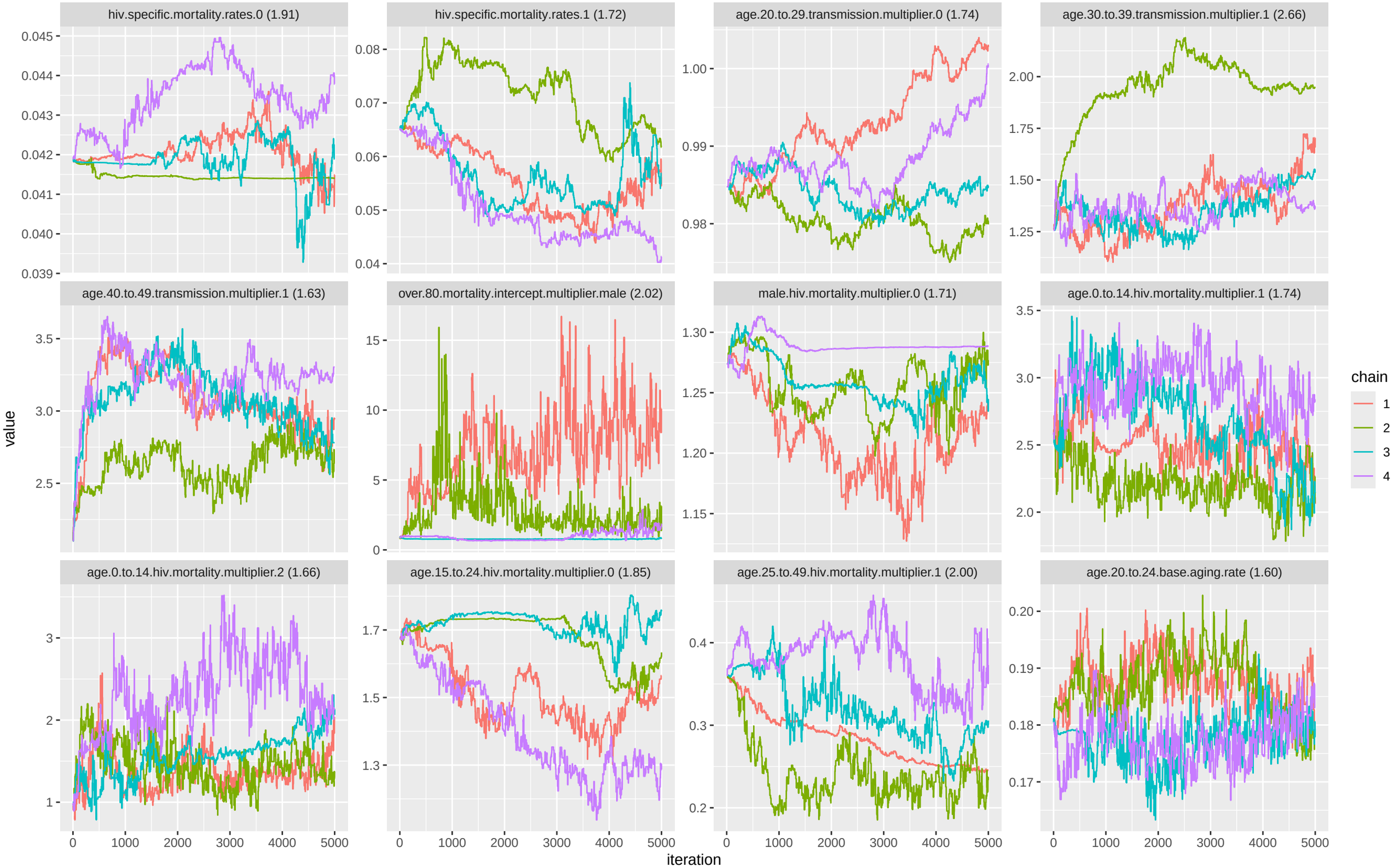
0
25000
50000
75000
100000

iteration

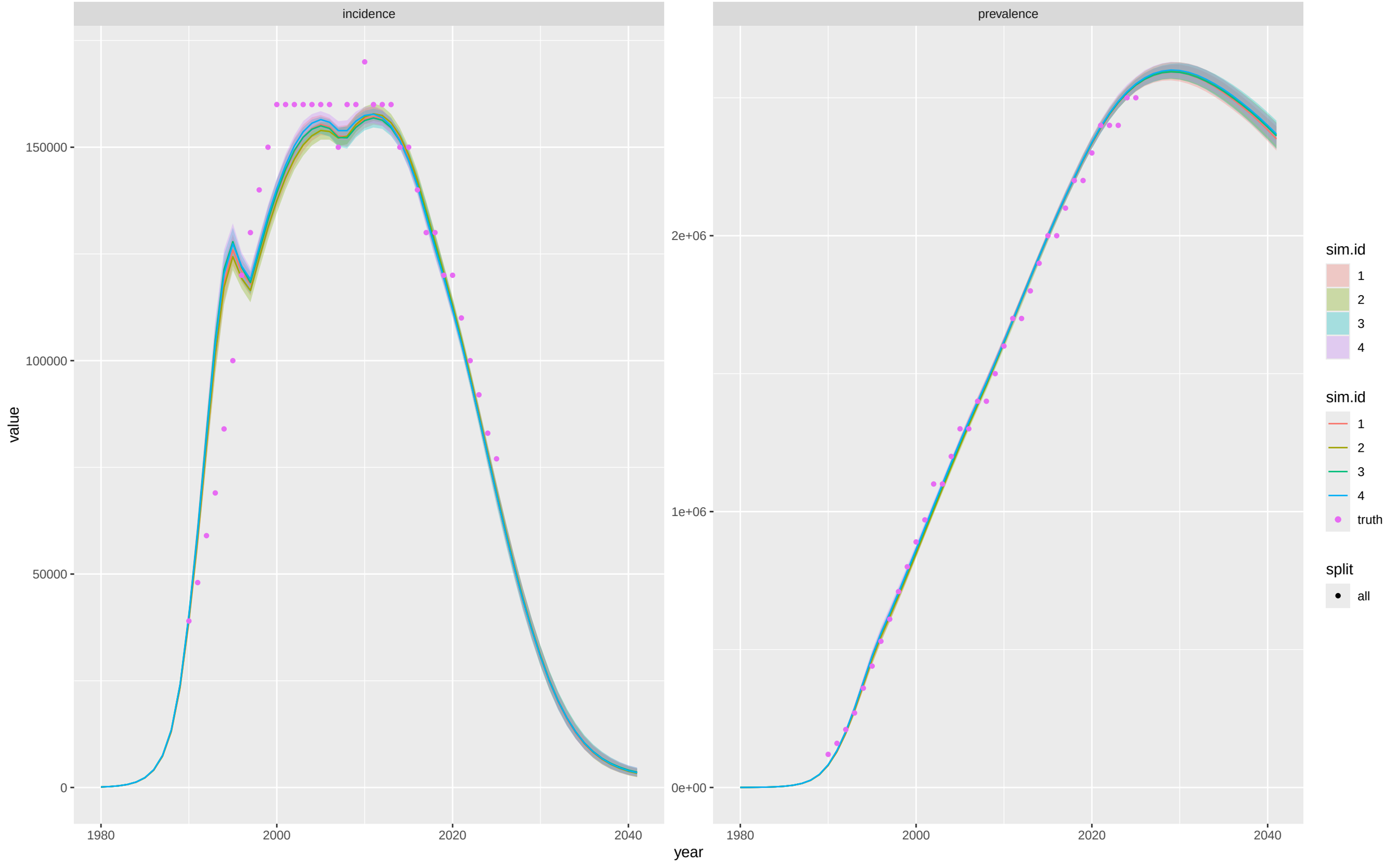
chain
1
2
3
4

type
Log Likelihood
Log Prior
Log Prior x Likelihood

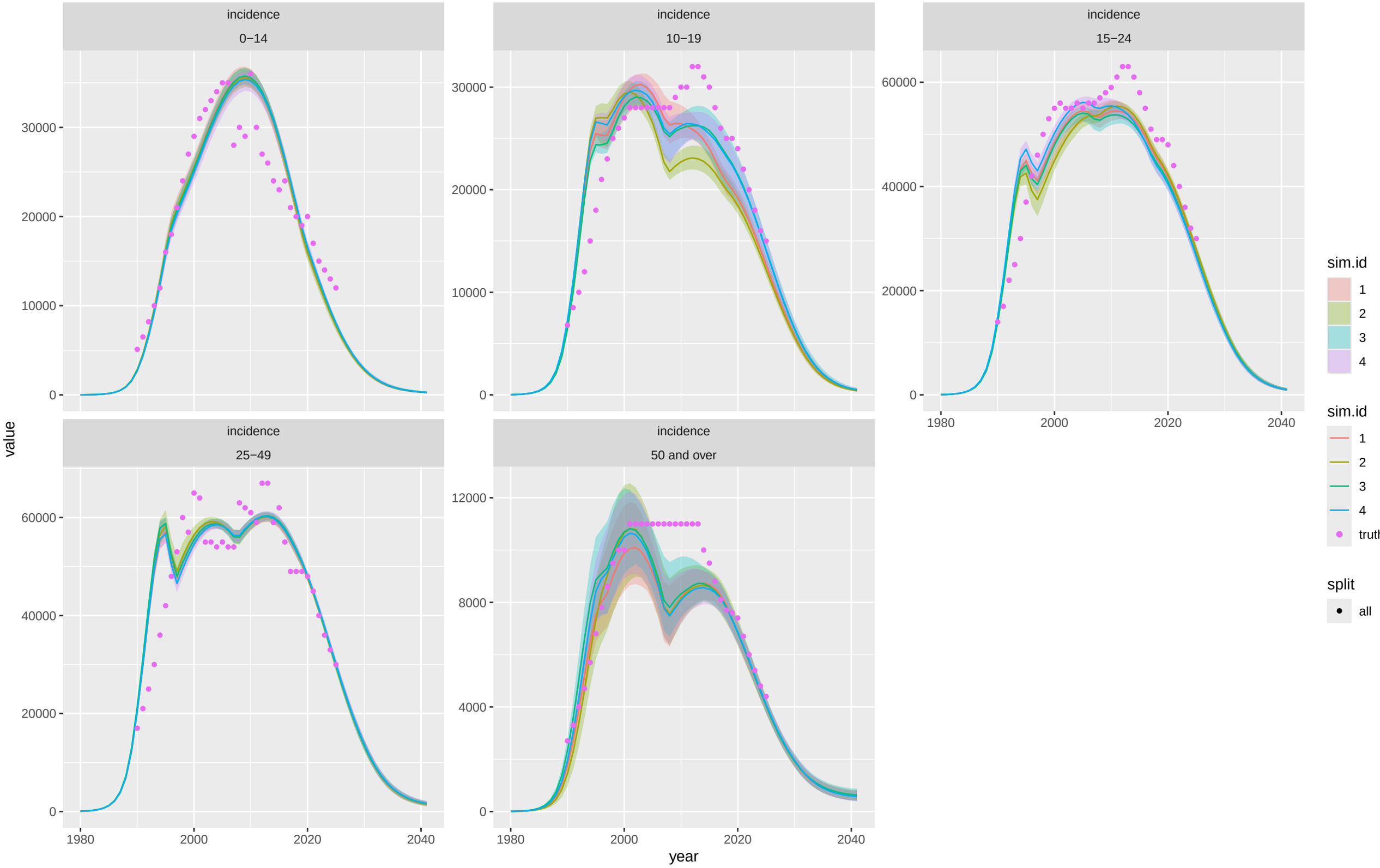
Mozambique step 2: trace



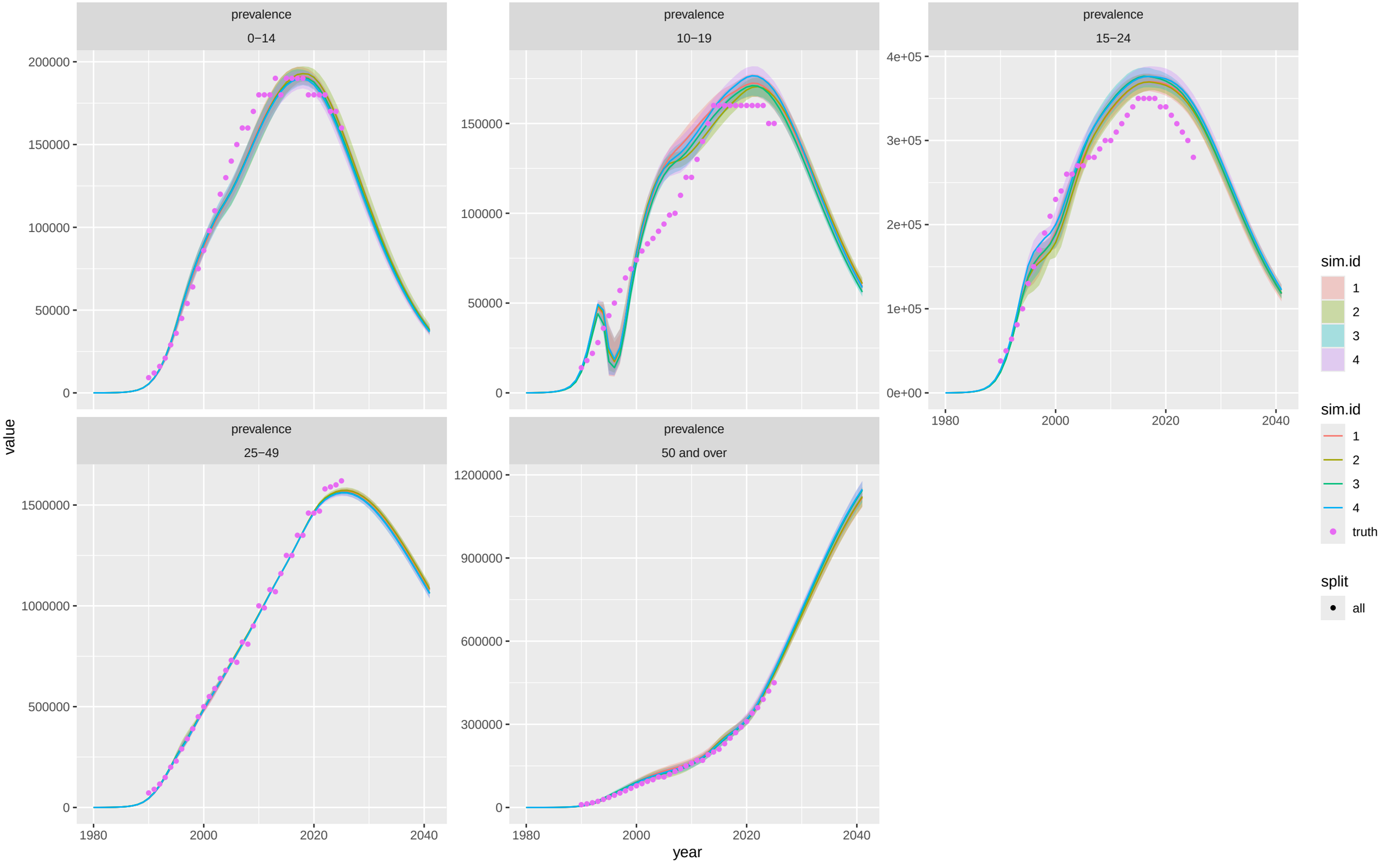
Mozambique step 2: fit (one color per chain)



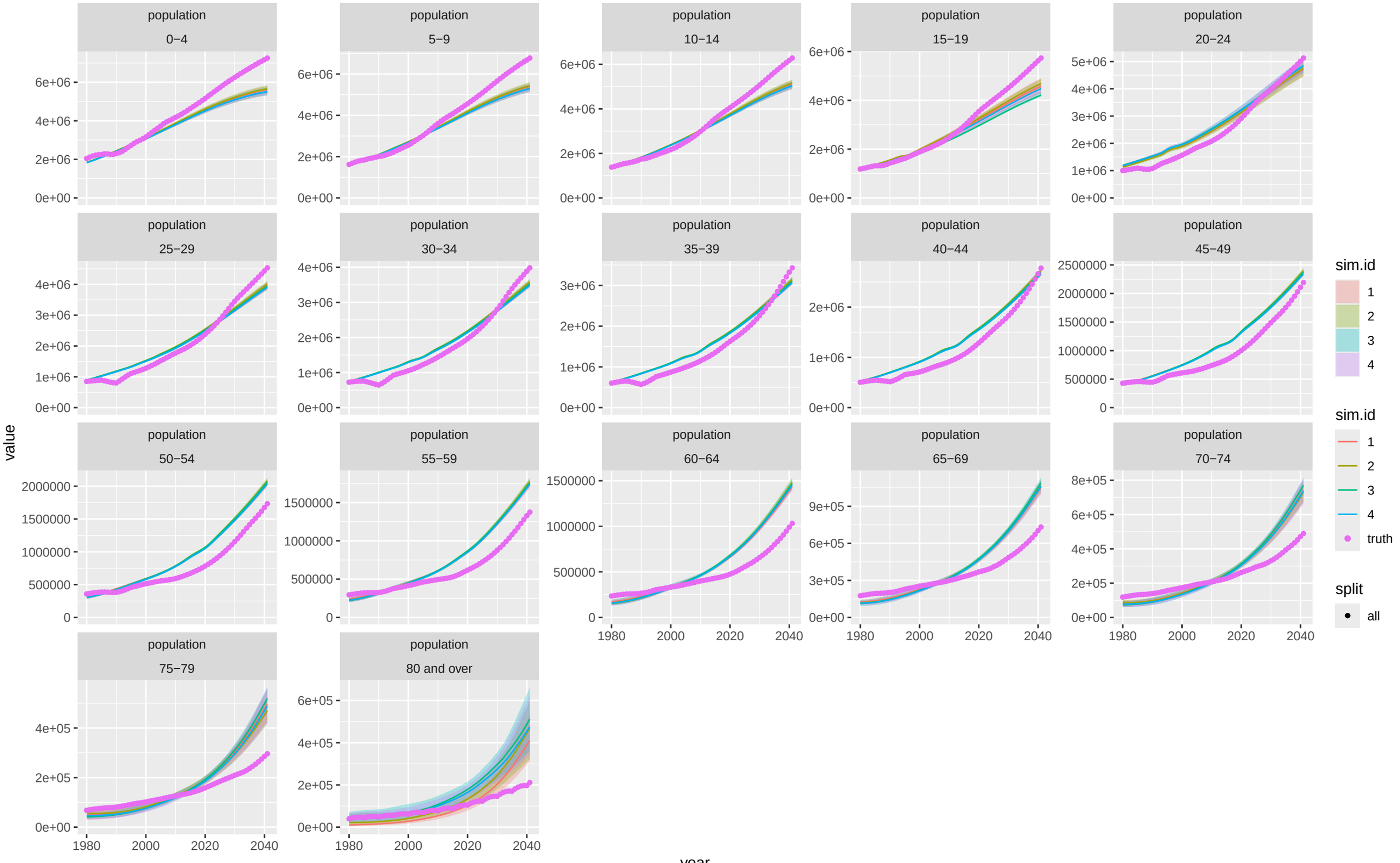
Mozambique step 2: incidence.by.age (one color per chain)



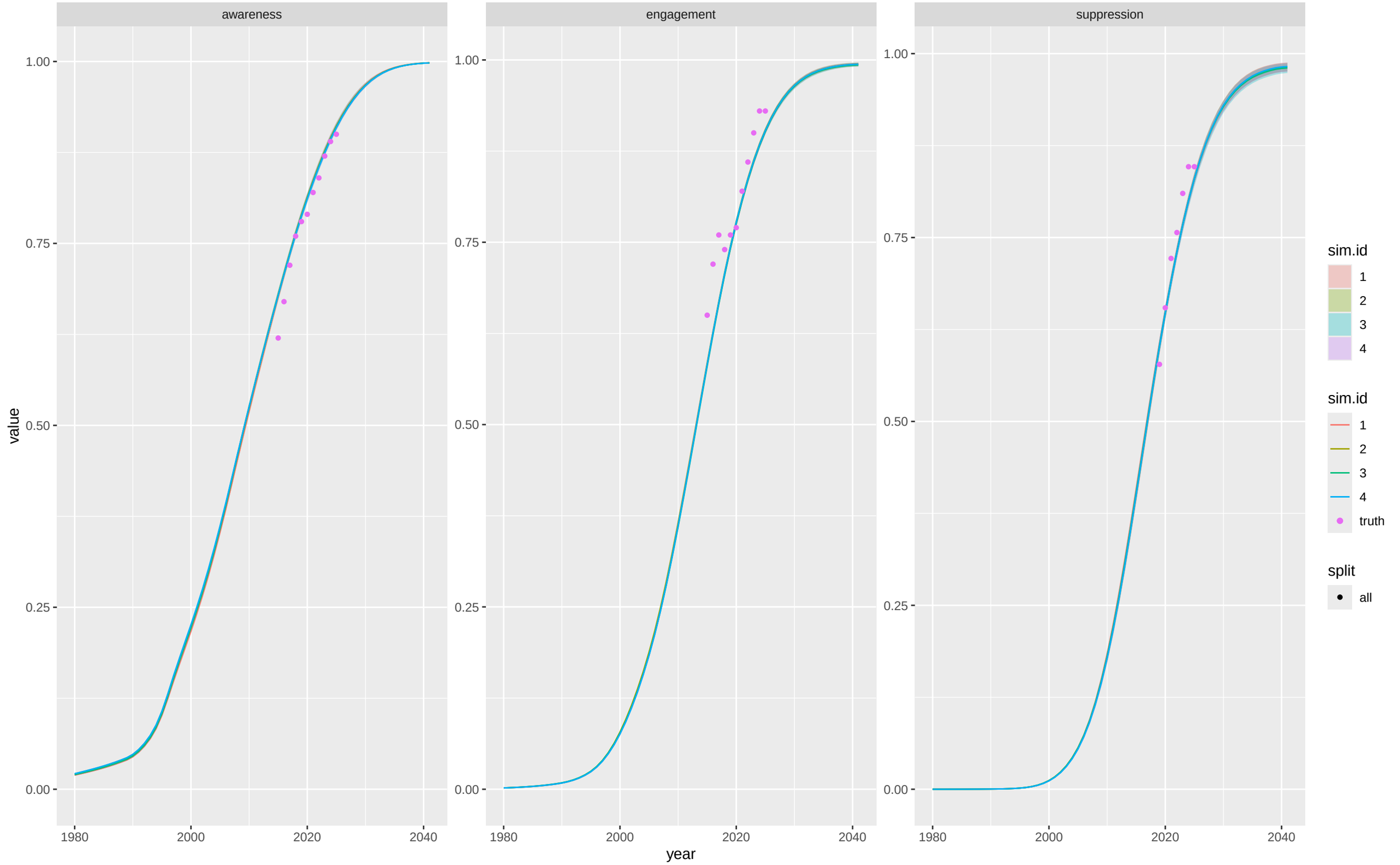
Mozambique step 2: prevalence.by.age (one color per chain)



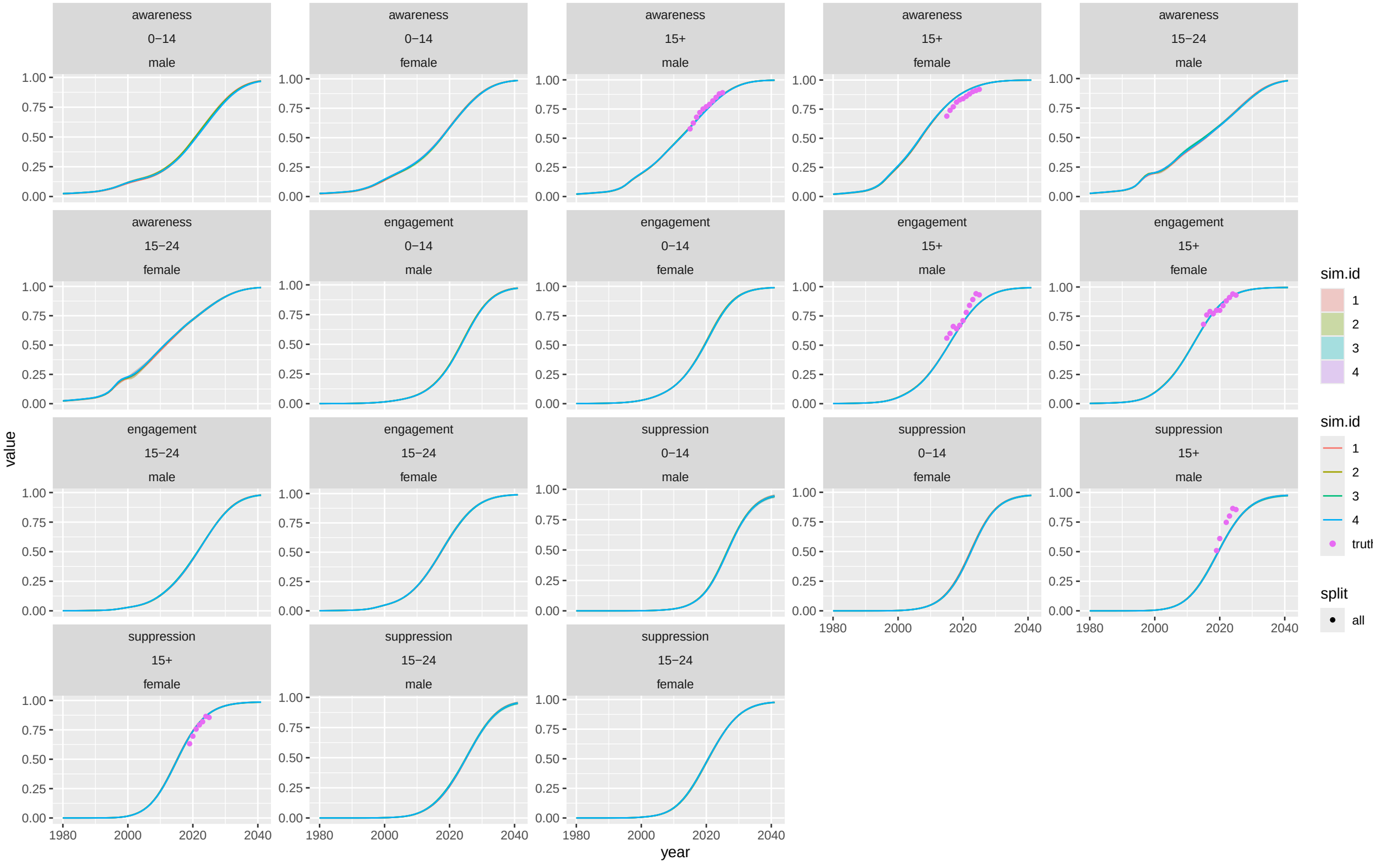
Mozambique step 2: population.by.age (one color per chain)



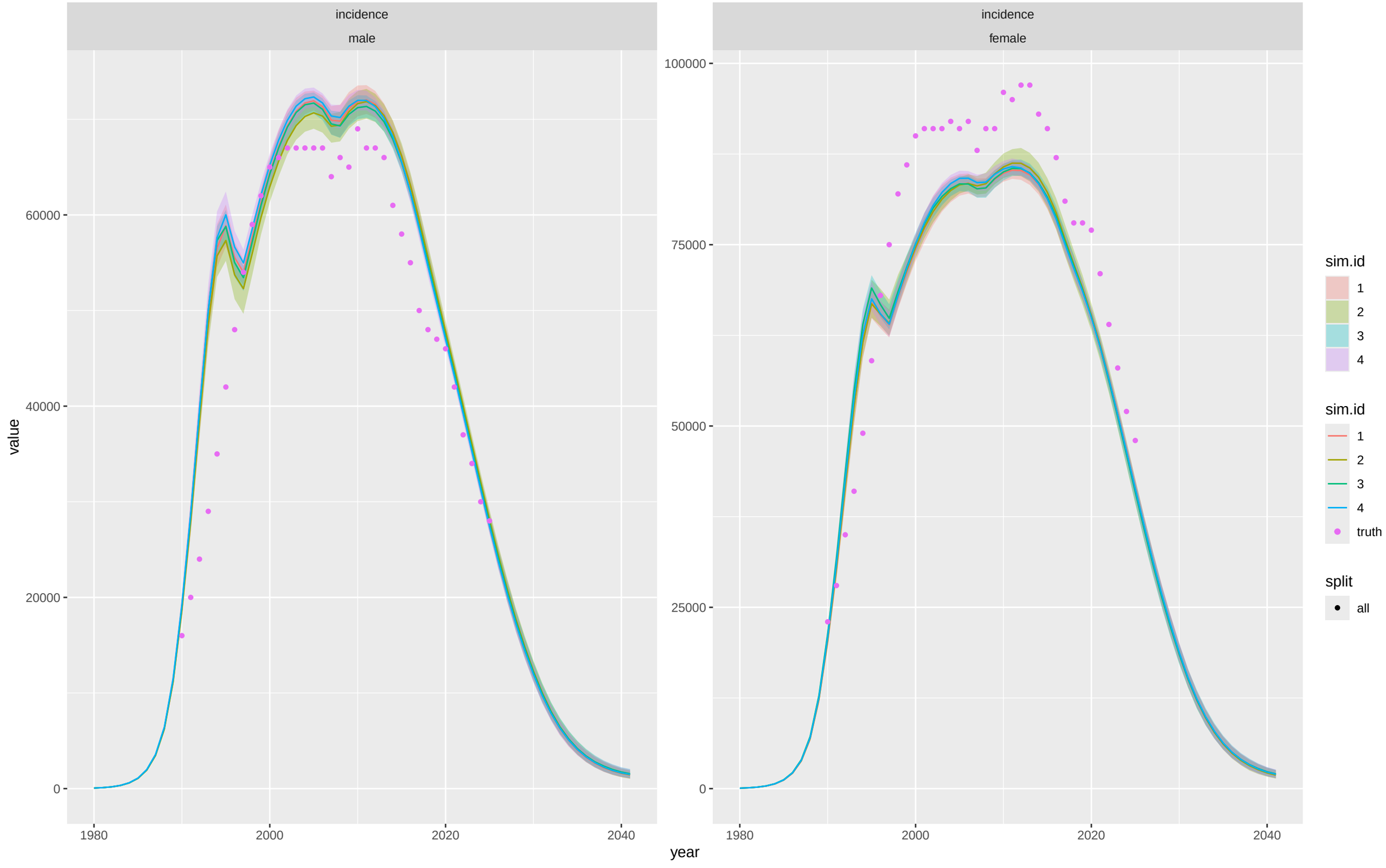
Mozambique step 2: cascade (one color per chain)



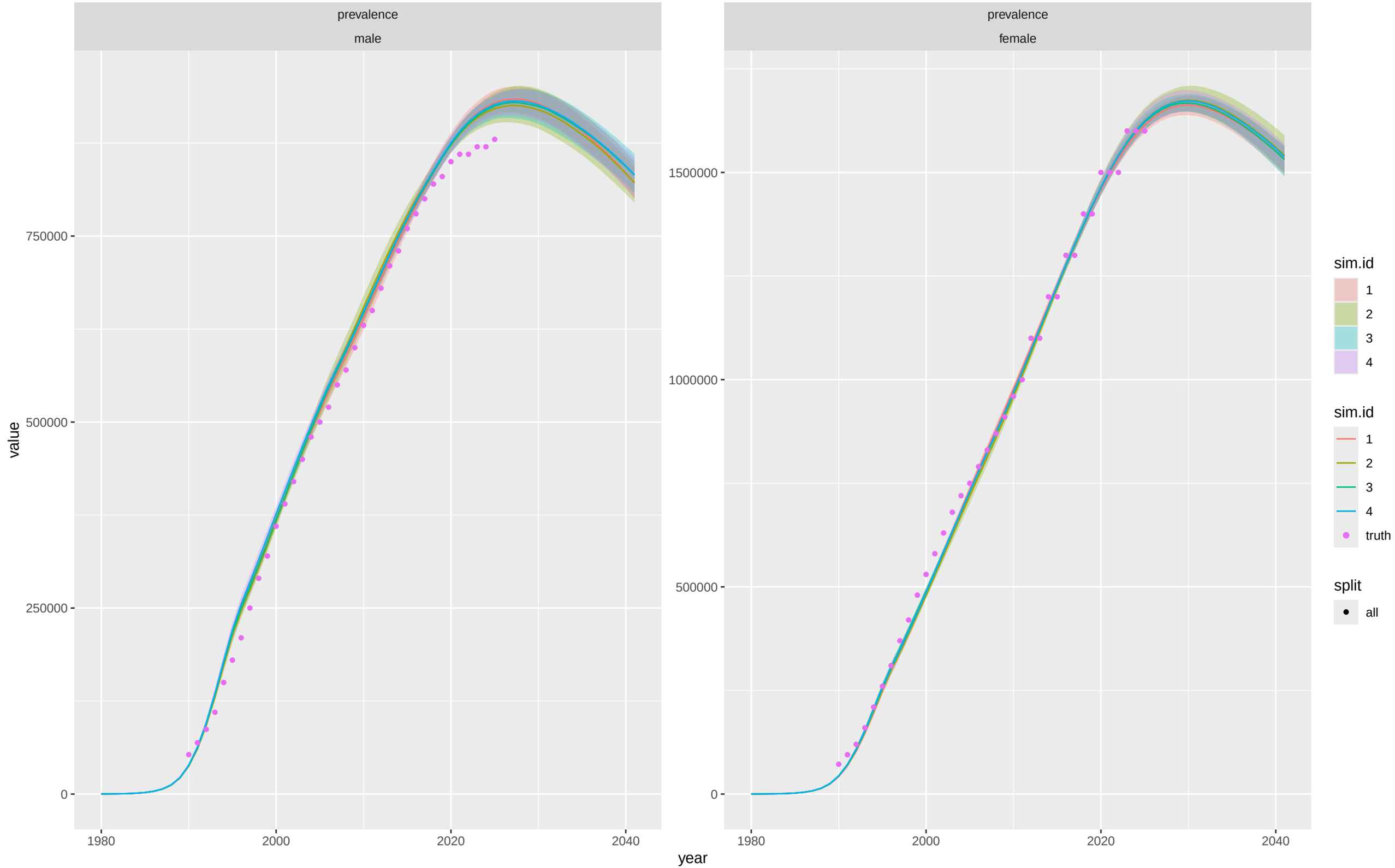
Mozambique step 2: cascade.by.age.sex (one color per chain)



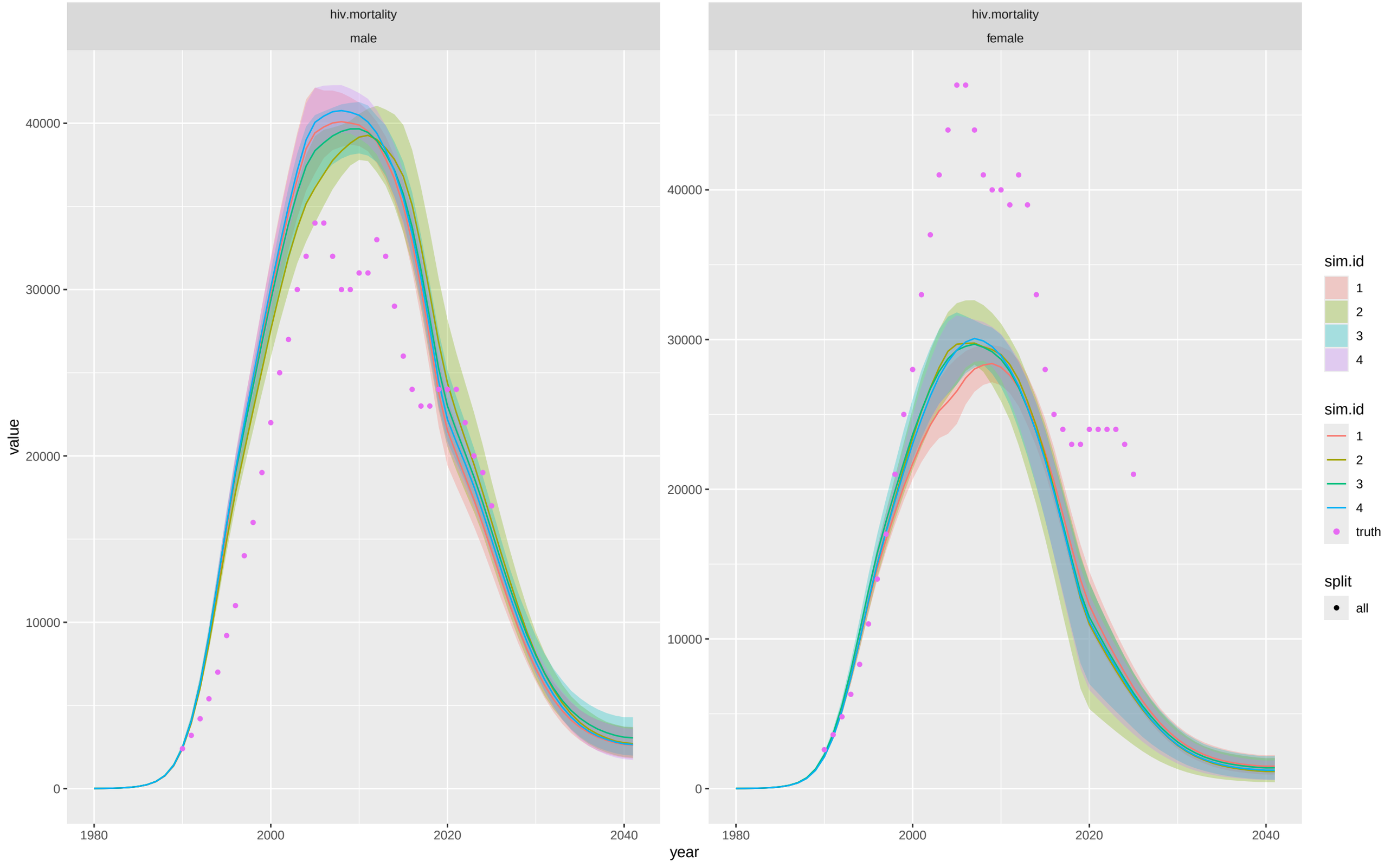
Mozambique step 2: incidence.by.sex (one color per chain)



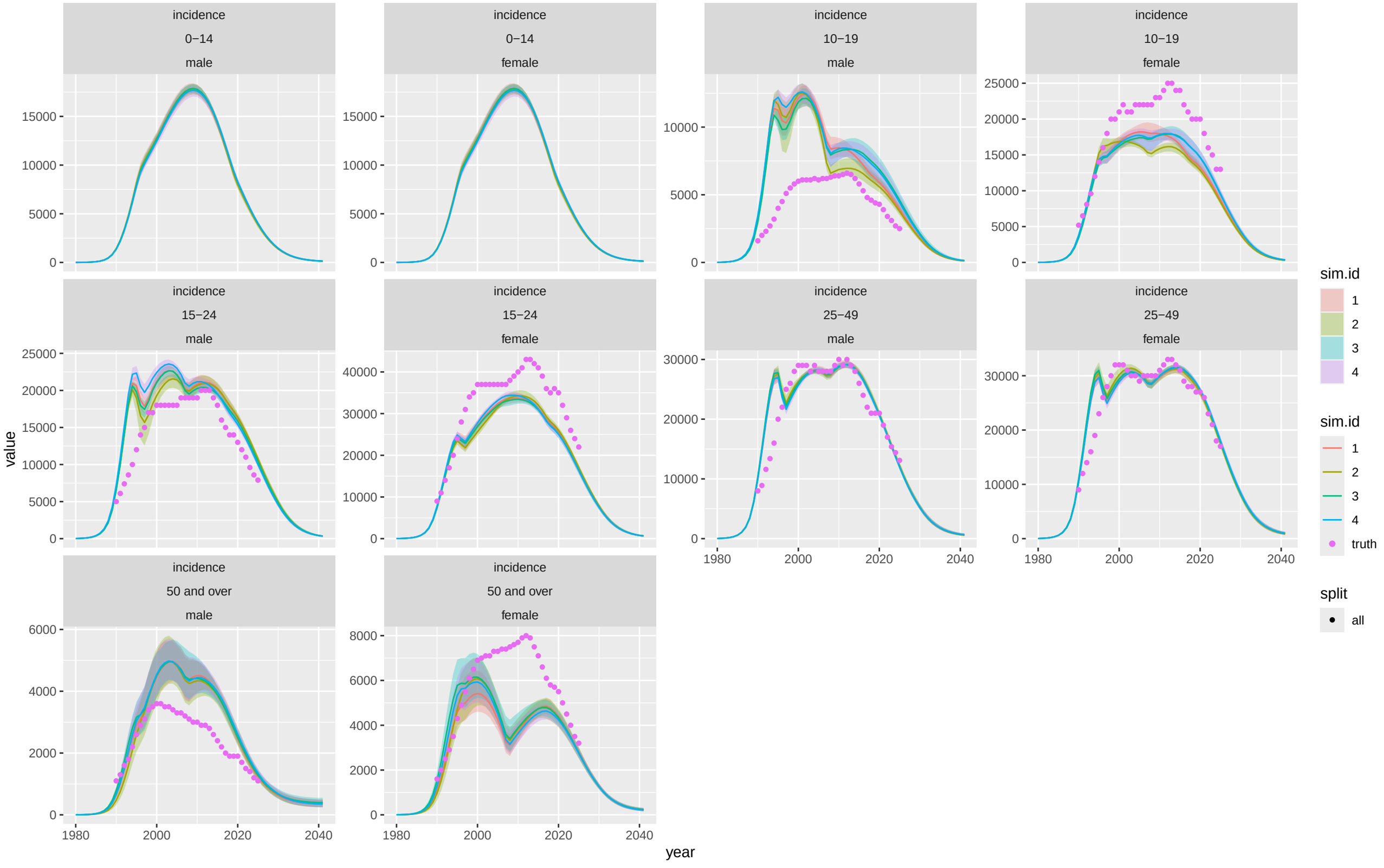
Mozambique step 2: prevalence.by.sex (one color per chain)



Mozambique step 2: hiv.mortality.by.sex (one color per chain)



Mozambique step 2: incidence.by.age.sex (one color per chain)



Mozambique step 2: prevalence.by.age.sex (one color per chain)

prevalence
0–14
male

prevalence
0–14
female

prevalence
10–19
male

prevalence
10–19
female

prevalence
15–24
male

prevalence
15–24
female

prevalence
25–49
male

prevalence
25–49
female

prevalence
50 and over
male

prevalence
50 and over
female

value

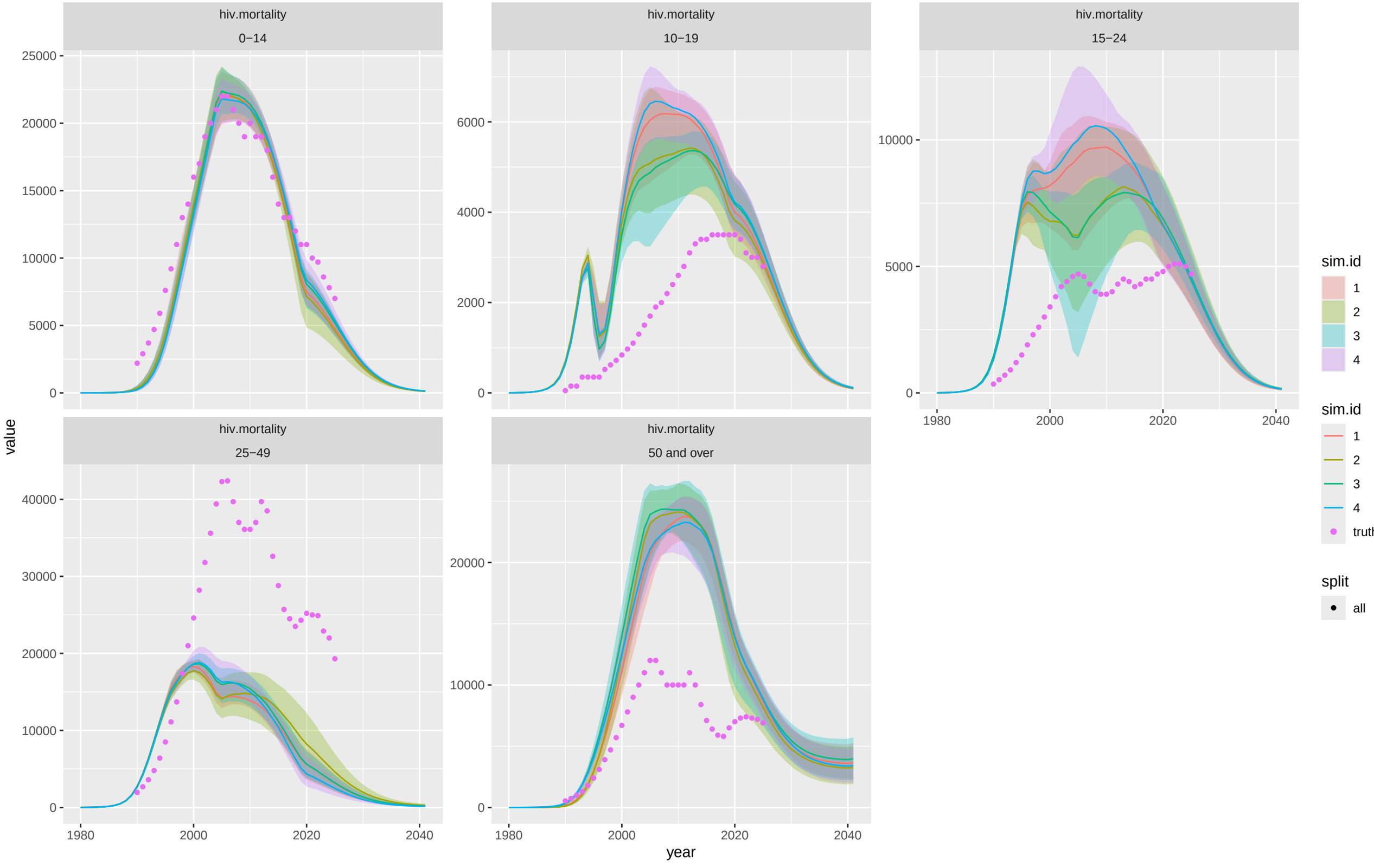
year

sim.id
1
2
3
4

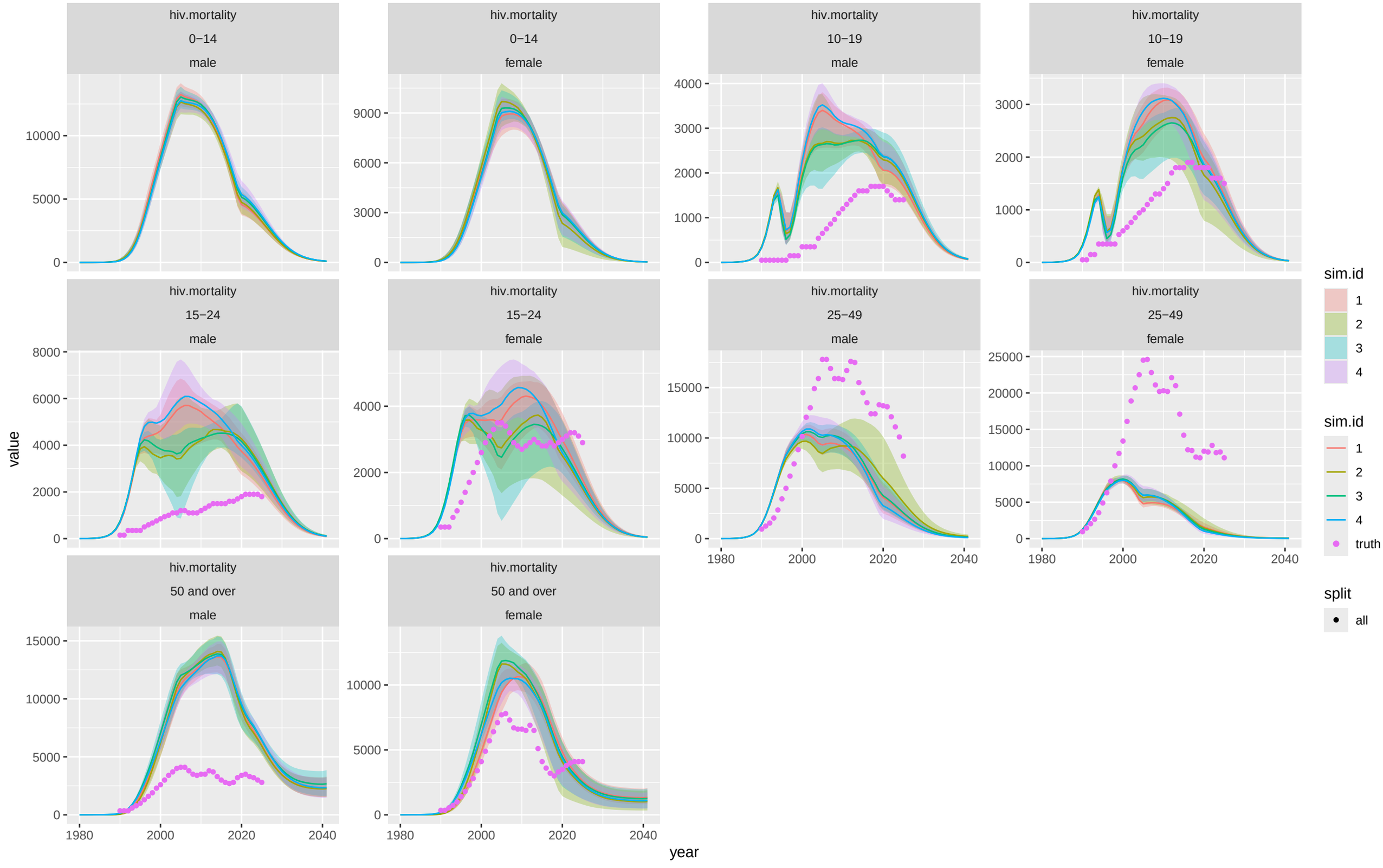
sim.id
1
2
3
4
truth

split
all

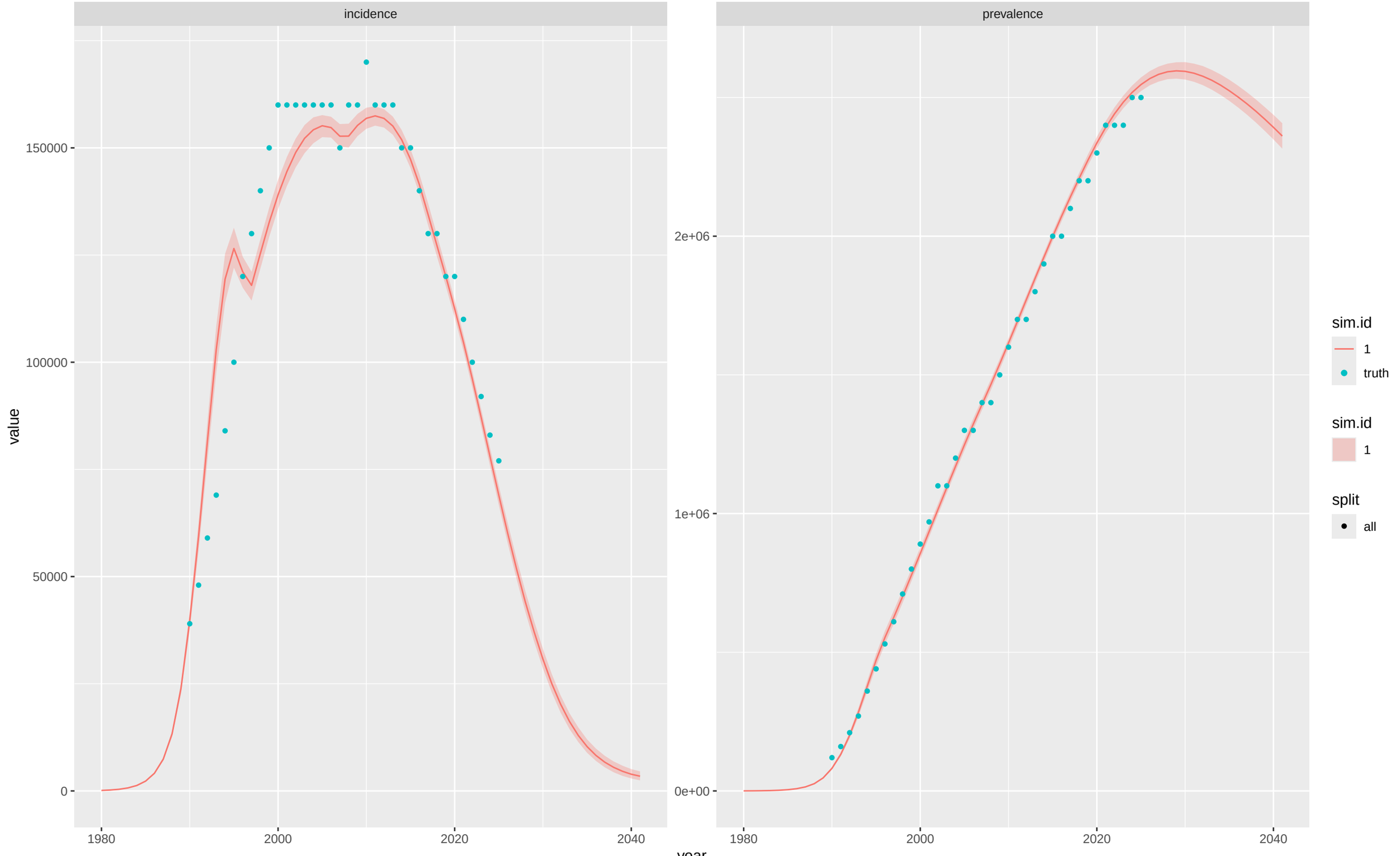
Mozambique step 2: hiv.mortality.by.age (one color per chain)



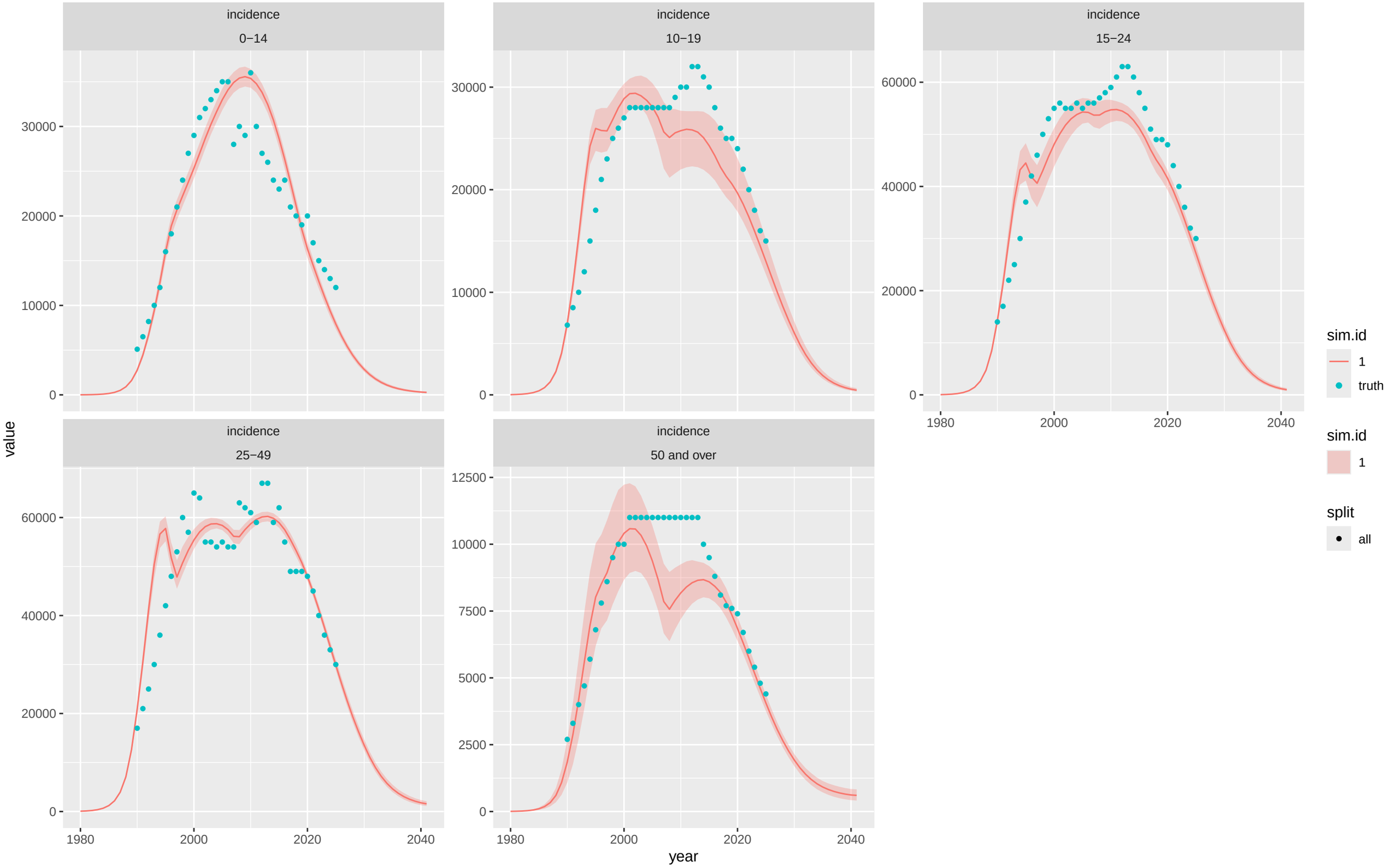
Mozambique step 2: hiv.mortality.by.age.sex (one color per chain)



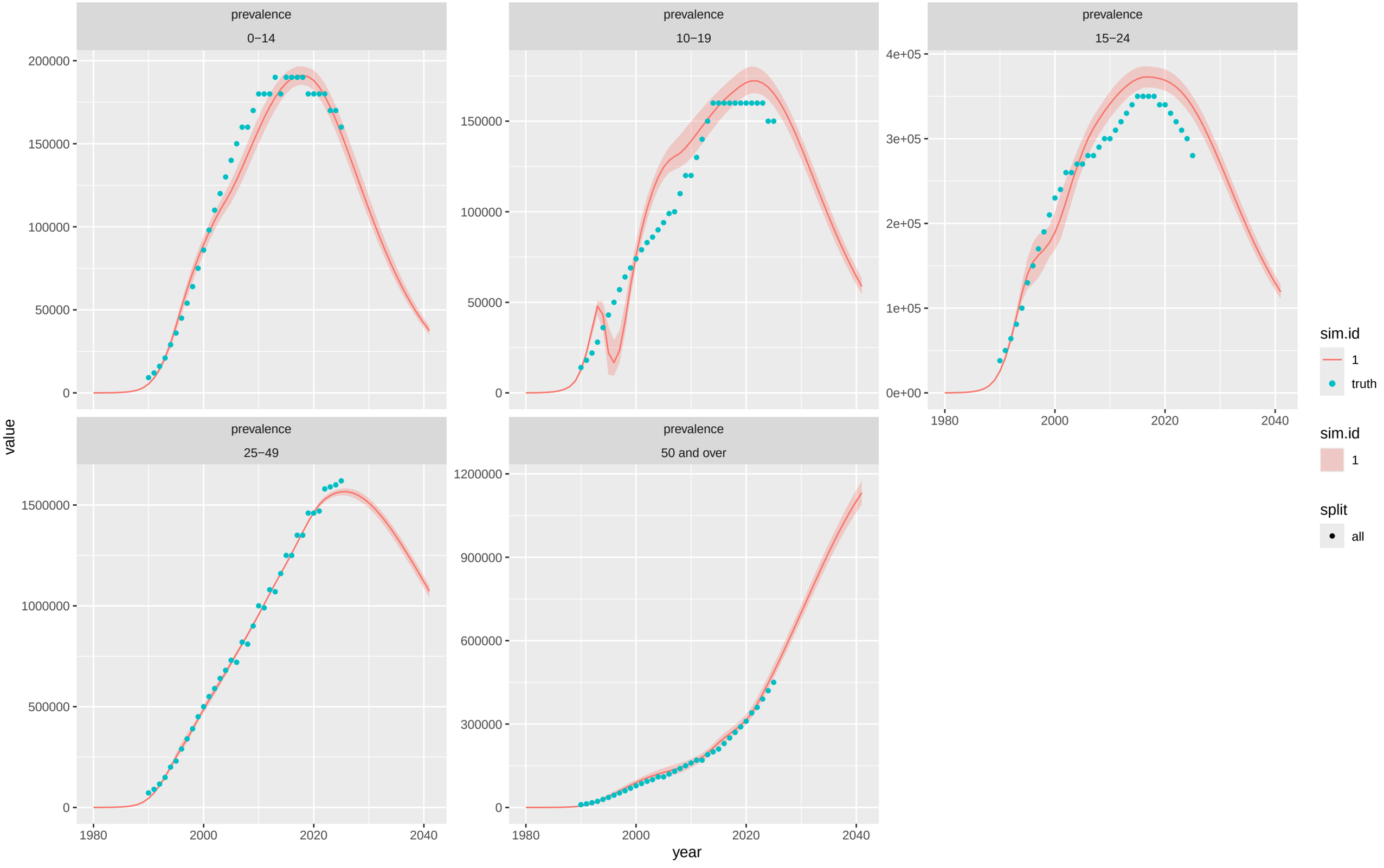
Mozambique: fit (merged Step 2 simset)



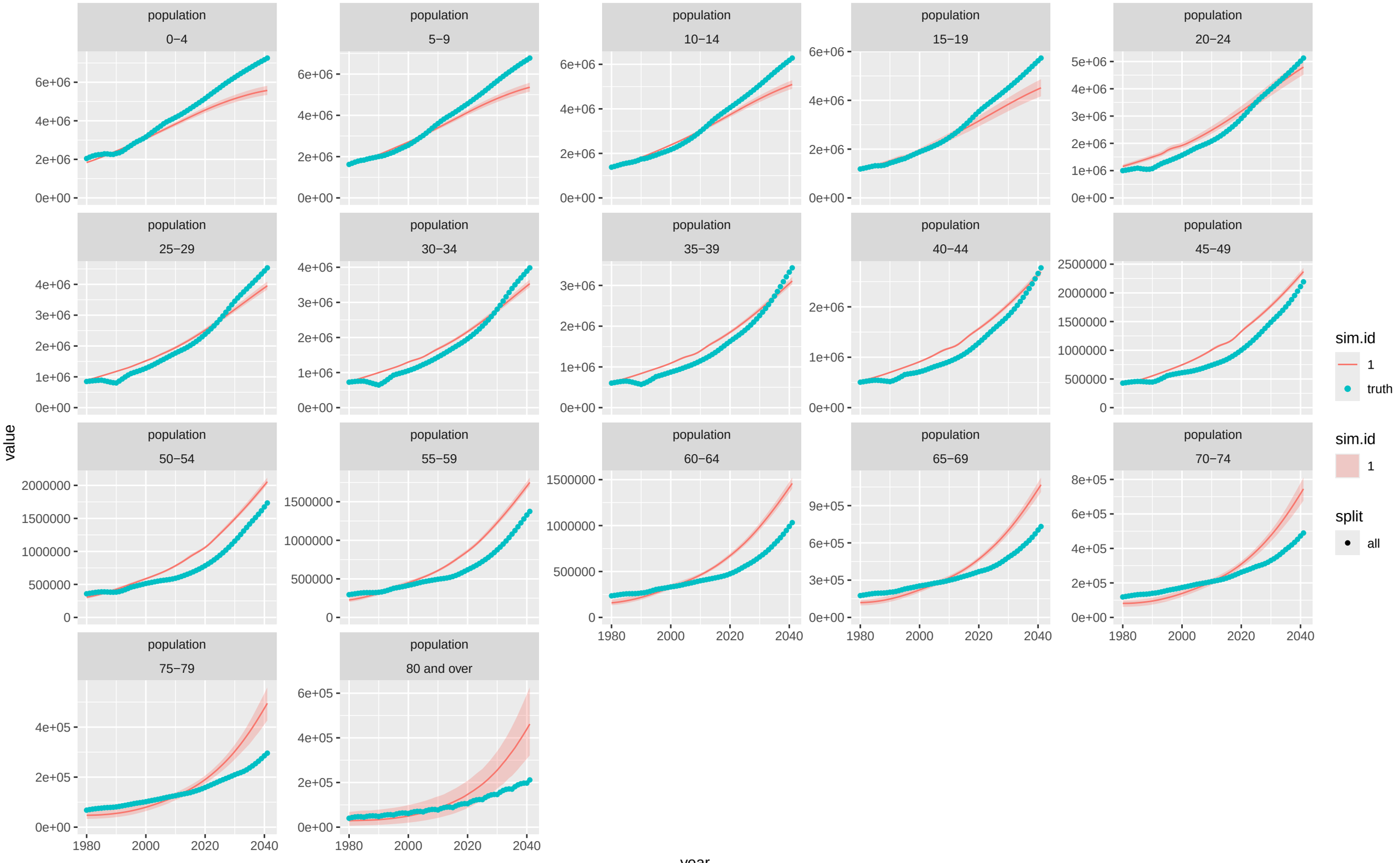
Mozambique: incidence.by.age (merged Step 2 simset)



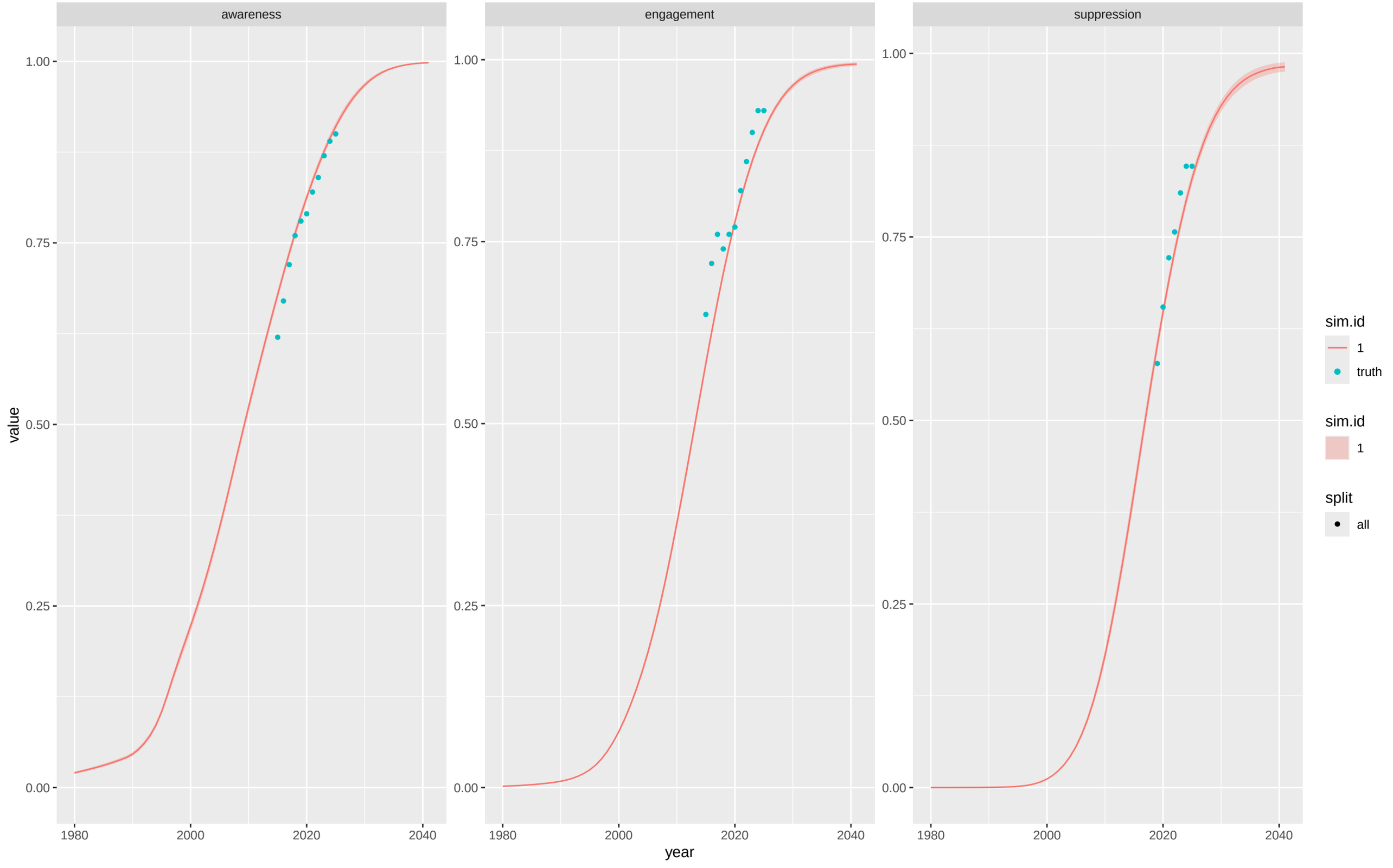
Mozambique: prevalence.by.age (merged Step 2 simset)



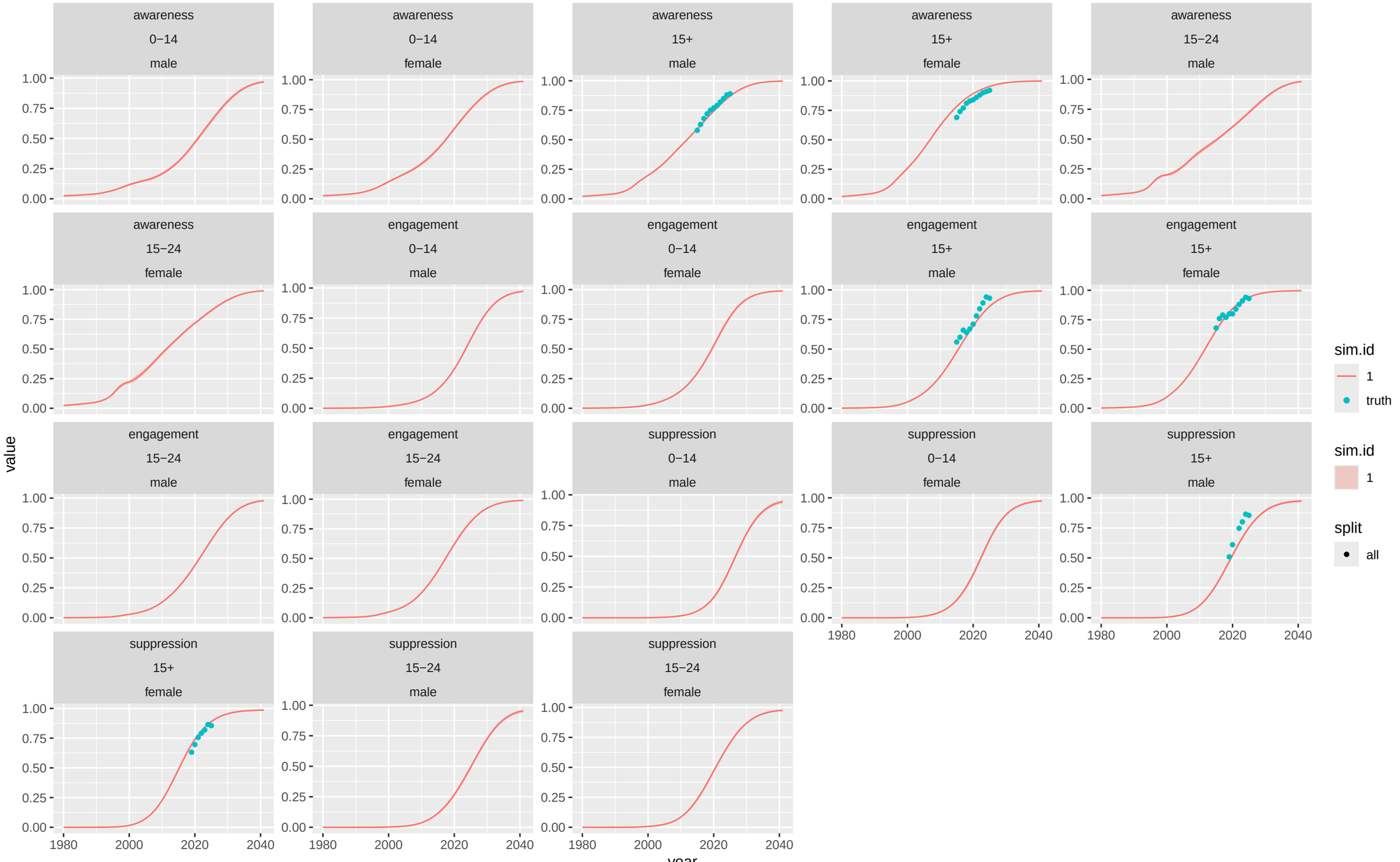
Mozambique: population.by.age (merged Step 2 simset)



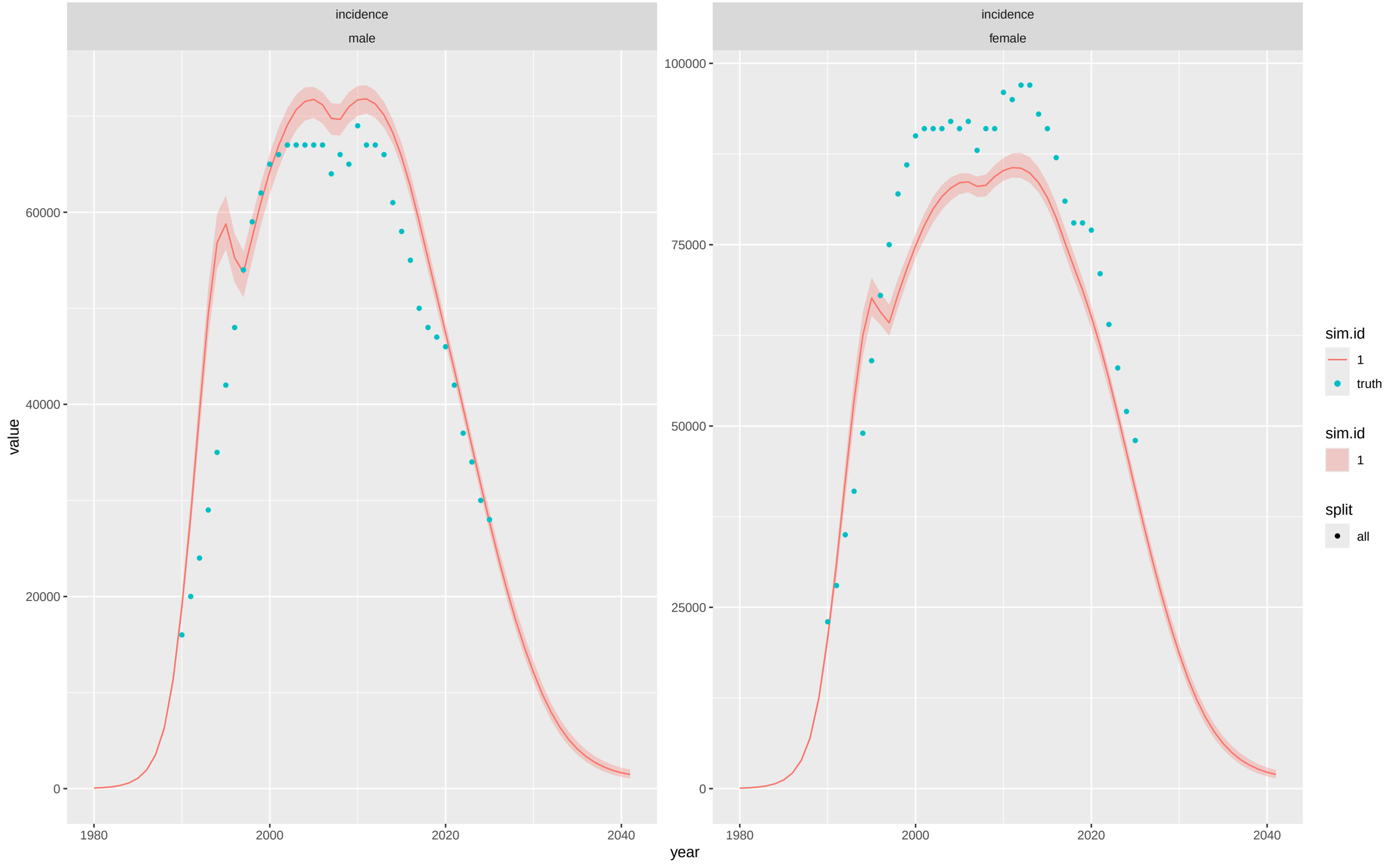
Mozambique: cascade (merged Step 2 simset)



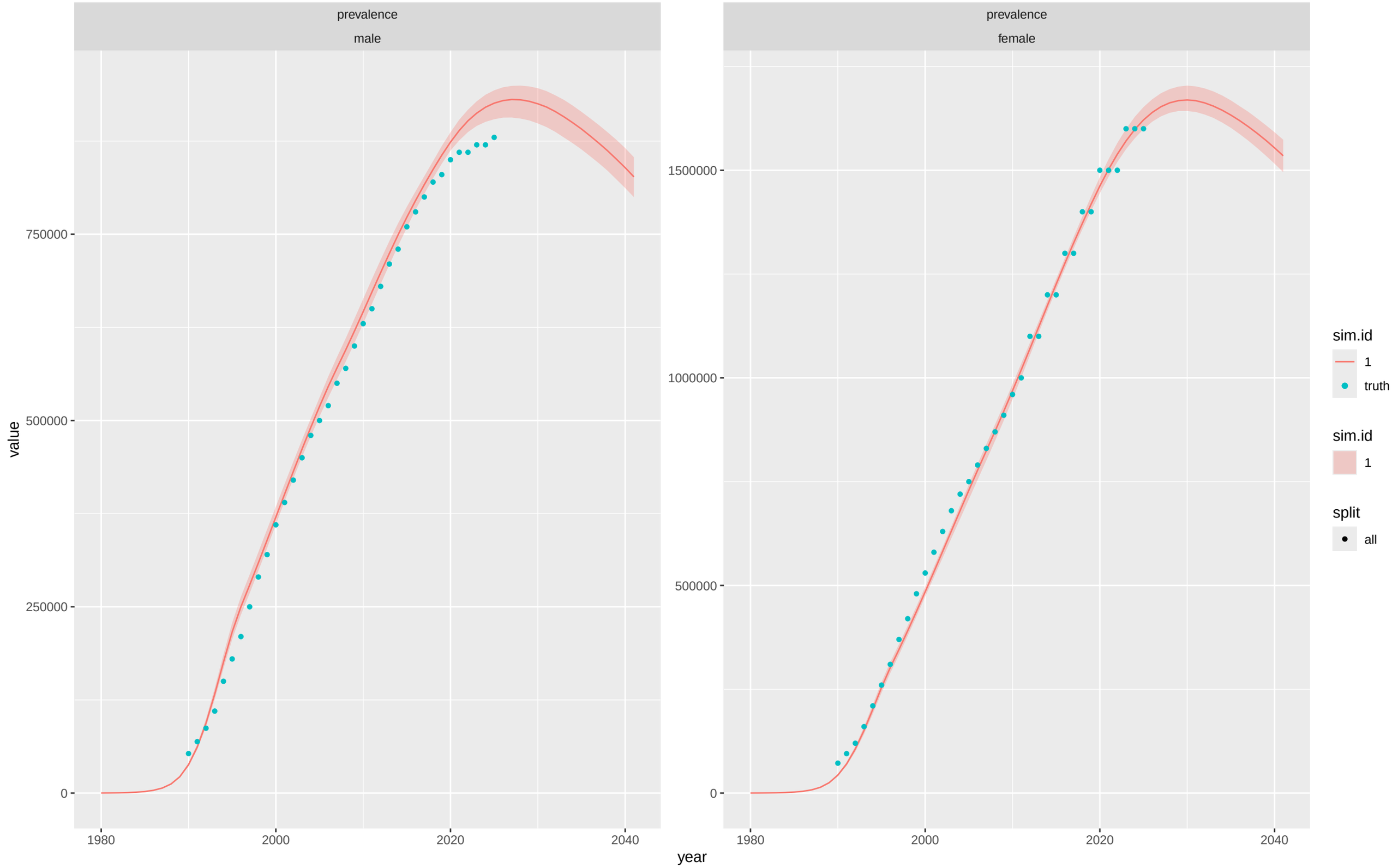
Mozambique: cascade.by.age.sex (merged Step 2 simset)



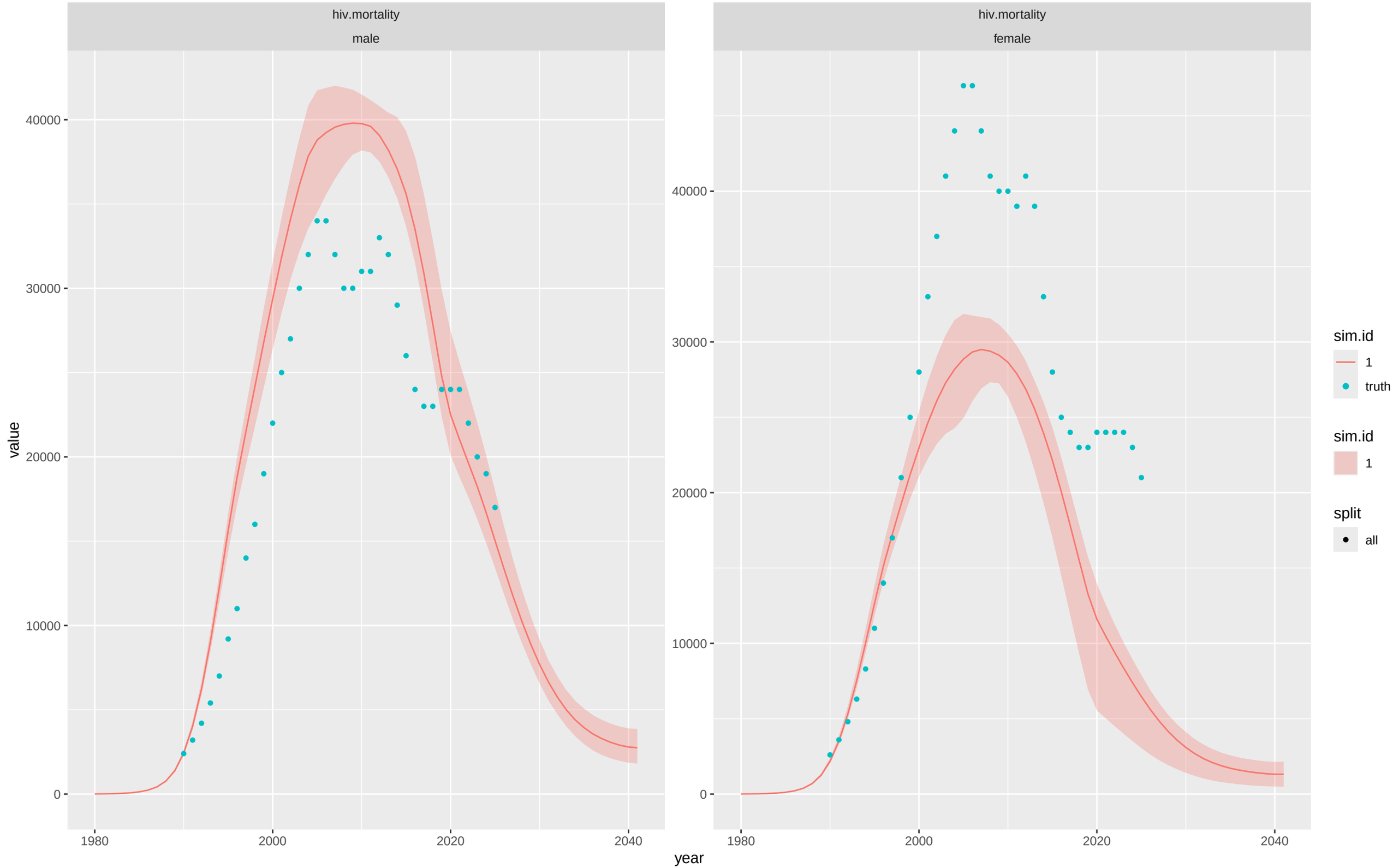
Mozambique: incidence.by.sex (merged Step 2 simset)



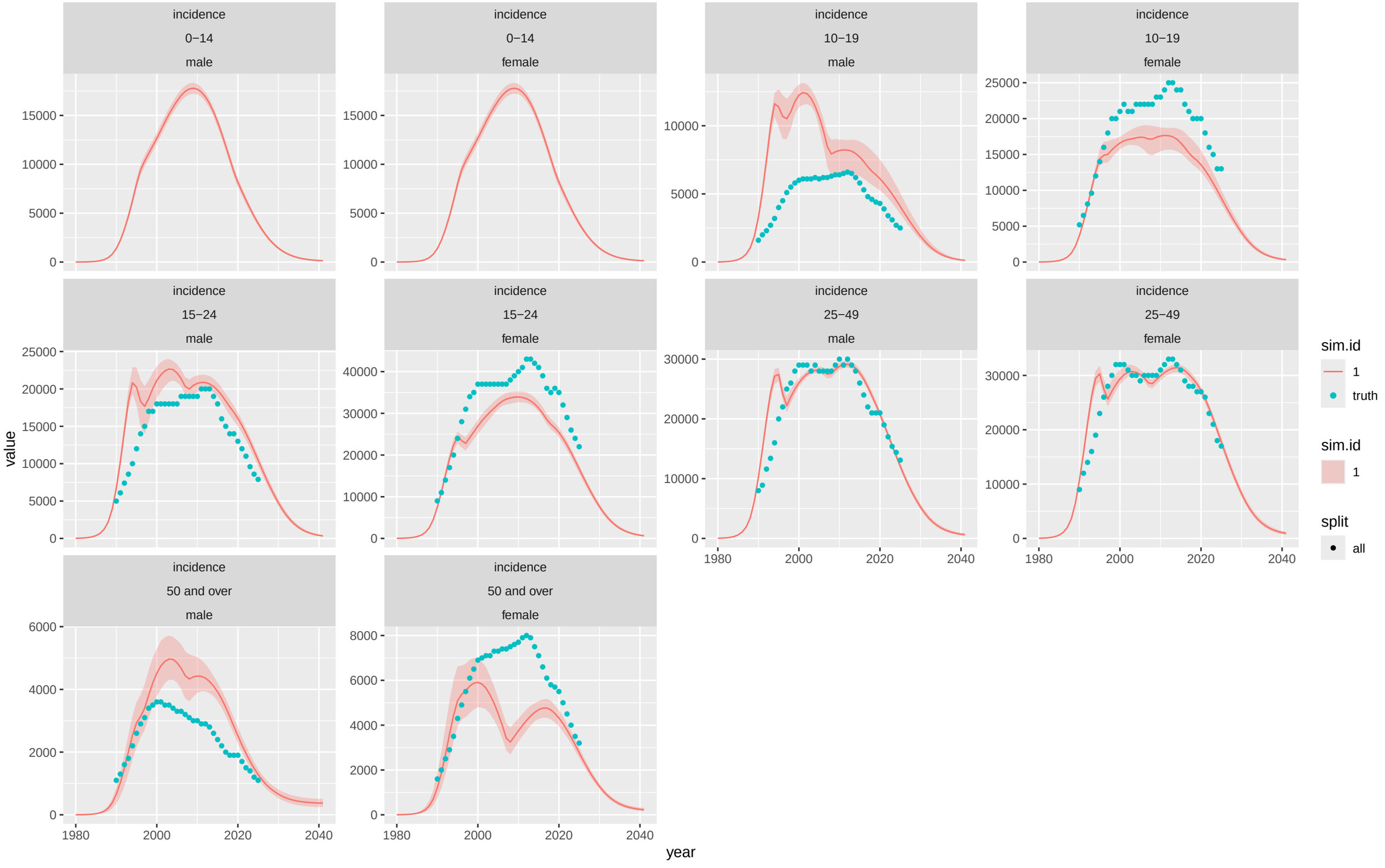
Mozambique: prevalence.by.sex (merged Step 2 simset)



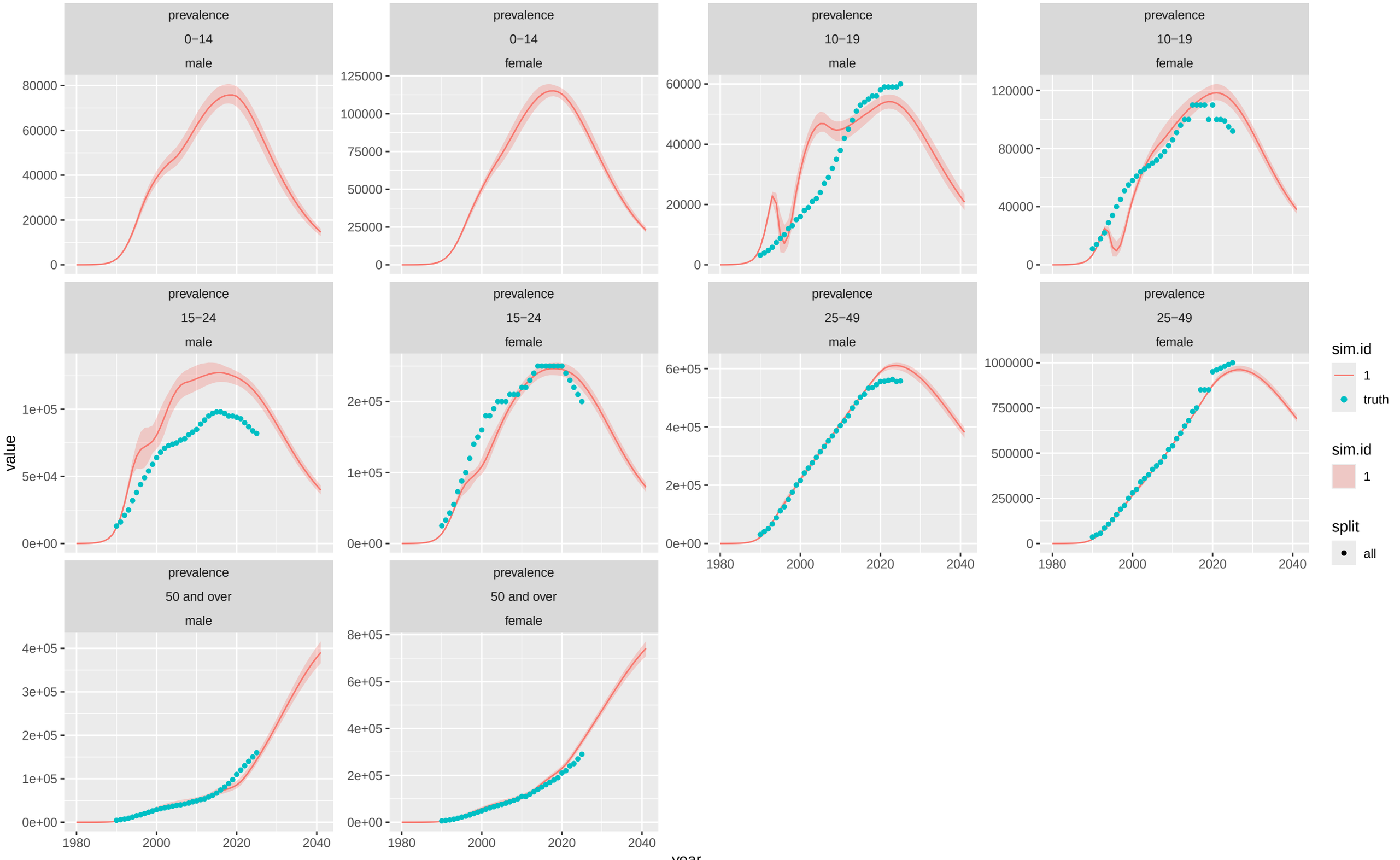
Mozambique: hiv.mortality.by.sex (merged Step 2 simset)



Mozambique: incidence.by.age.sex (merged Step 2 simset)



Mozambique: prevalence.by.age.sex (merged Step 2 simset)



Mozambique: hiv.mortality.by.age (merged Step 2 simset)



Mozambique: hiv.mortality.by.age.sex (merged Step 2 simset)

